

Analytical MCQ Assessment

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. A network administrator wants to block incoming traffic based solely on source IP address and port number without tracking the state of active connections. Which type of firewall is most suitable for this task? (1 Marks)

- A. Stateful Inspection Firewall
- B. Packet Filtering Firewall
- C. Application-Level Gateway
- D. Next-Generation Firewall

Q2. Which type of firewall operates at the Session layer of the OSI model to monitor TCP handshaking and determine if a requested session is legitimate? (1 Marks)

- A. Packet Filtering Firewall
- B. Circuit-Level Gateway
- C. Application-Level Gateway
- D. Stateful Inspection Firewall

Q3. An organization requires a firewall that can inspect the actual content of HTTP requests to block specific SQL injection patterns. Which traditional firewall type is best suited for this? (1 Marks)

- A. Packet Filter
- B. Circuit-Level Gateway
- C. Application-Level Gateway (Proxy)
- D. Stateful Inspection Firewall

Q4. How does a stateful inspection firewall determine whether to allow an incoming packet that does not match an explicit inbound rule? (1 Marks)

- A. It checks if the packet is part of an established outbound connection in its state table.
- B. It automatically routes the packet to a honeypot for analysis.
- C. It performs deep packet inspection on the payload.
- D. It prompts the administrator for manual approval in real-time.

Q5. What primary capability distinguishes a Next-Generation Firewall (NGFW) from a traditional stateful inspection firewall? (1 Marks)

- A. Ability to filter packets based on source and destination IP addresses.
- B. Integration of deep packet inspection, intrusion prevention (IPS), and application awareness.
- C. Operation strictly at the Network and Transport layers of the OSI model.
- D. Lower latency by avoiding payload inspection.

Q6. A company needs to protect its public-facing web servers specifically from OWASP

Top 10 vulnerabilities like Cross-Site Scripting (XSS). Which specialized firewall should they deploy? (1 Marks)

- A. Stateful Network Firewall
- B. Web Application Firewall (WAF)
- C. Circuit-Level Gateway
- D. Packet Filtering Router

Q7. Which scenario best describes the deployment of a host-based firewall? (1 Marks)

- A. A hardware appliance protecting an entire corporate local area network.
- B. A software application running on an individual server to control traffic to and from that specific machine.
- C. A cloud-based service filtering traffic for multiple remote branch offices.
- D. A physical device placed between the internal switch and the external router.

Q8. In a screened-subnet (DMZ) architecture, where is the external-facing firewall typically positioned? (1 Marks)

- A. Between the internal private network and the DMZ.
- B. Directly on the database server hosting sensitive customer data.
- C. Between the public Internet and the DMZ.
- D. Inside the local active directory domain controller.

Q9. Which firewall technology hides internal IP addresses by translating them into a single public IP address for outbound traffic? (1 Marks)

- A. Deep Packet Inspection
- B. Network Address Translation (NAT)
- C. Packet Switching
- D. Stateful Packet Filtering

Q10. Which of the following security threats is a traditional packet-filtering firewall least effective at preventing? (1 Marks)

- A. IP spoofing attacks from external networks.
- B. Unauthorized access to restricted TCP ports.
- C. An exploit payload hidden inside a legitimate HTTP POST request.
- D. Basic denial-of-service attacks targeting network bandwidth.

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B

Rationale: Packet filtering firewalls inspect packets individually based on header information like IP addresses and ports without tracking connection state.

Q2. Correct Answer: B

Rationale: Circuit-level gateways verify the TCP handshake at the Session layer without inspecting the actual packet payloads.

Q3. Correct Answer: C

Rationale: Application-level gateways inspect the payload of packets at the Application layer to filter specific command and content types.

Q4. Correct Answer: A

Rationale: Stateful firewalls keep track of active connections in a state table and automatically allow returning traffic associated with those connections.

Q5. Correct Answer: B

Rationale: NGFWs combine traditional firewall capabilities with deep packet inspection, intrusion prevention systems, and application-level control.

Q6. Correct Answer: B

Rationale: A WAF is specifically designed to analyze and filter HTTP/HTTPS traffic to protect web applications from application-layer attacks.

Q7. Correct Answer: B

Rationale: Host-based firewalls are software solutions installed directly on individual endpoints to secure only that specific host.

Q8. Correct Answer: C

Rationale: The external firewall sits between the untrusted public Internet and the DMZ to filter initial incoming traffic.

Q9. Correct Answer: B

Rationale: NAT is commonly integrated into firewalls to map private IP addresses to public ones, masking the internal network structure.

Q10. Correct Answer: C

Rationale: Traditional packet filters only inspect headers and cannot detect malicious payloads hidden within legitimate application-layer traffic.